KS3: INDUSTRIALISATION, EMPIRE, AND POWER (1750–1900)

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COURSE TEXTBOOK & READING LOG

Enquiry: *Industrialisation, Empire, and Power: How did 19th-century Britain transform at home and abroad?*

Contents

L1: What powered the Industrial Revolution, and how did a Fareham ironmaster change the world?

LEARNING OBJECTIVES

- [ ] Understand the difference between the domestic system and the factory system.
- [ ] Explain the significance of Henry Cort's puddling and rolling processes.
- [ ] Evaluate the importance of location versus innovation in Cort's success.

Do Now Activity

1. What is meant by the term 'Divine Right of Kings'?

2. Who became King after the English Civil War ended and Oliver Cromwell died?

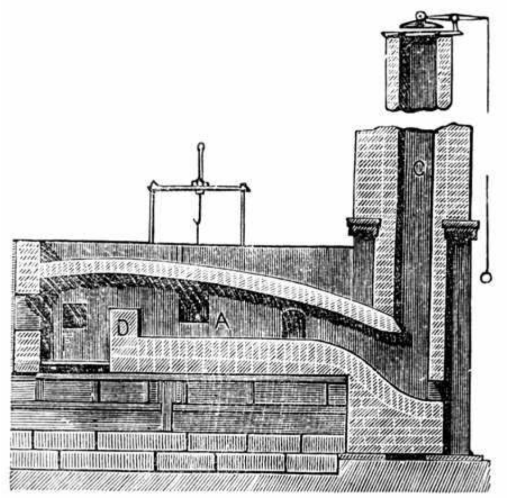
3. Why did the Gunpowder Plot of 1605 fail?

Vocabulary Check

Write a short paragraph using at least FOUR of the vocabulary words below correctly.

Industrial Revolution | Pig Iron | Wrought Iron | Puddling Process | Rolling Mill

Source A: The Puddling Furnace



A historical cross-section diagram of a puddling furnace, the revolutionary process Henry Cort pioneered at his Funtley ironworks in Hampshire.

Before the 1700s, Britain was an agricultural country. Most goods were made by hand in people's homes (the "domestic system"), and the only machines that existed were powered by natural, unreliable sources like wind, fast-flowing rivers, or the muscle power of horses. However, beneath Britain's soil lay massive reserves of **coal**—a fuel that burned much hotter than wood. In 1712, Thomas Newcomen invented the first practical steam engine, which used burning coal to pump water out of deep mines. Later, in 1769, an engineer named **James Watt** drastically improved the design, making the steam engine much more efficient and powerful. Watt's steam engine changed everything: it meant that massive factories no longer needed to be built next to rivers. They could be built anywhere, powered entirely by coal and steam. This shift from farming and hand-crafting to mass factory production driven by steam power is known as the **Industrial Revolution**. However, to build these enormous new steam engines, factories, railways, and bridges, Britain desperately needed one crucial material: **iron**.

Q1. Why was James Watt's improvement of the steam engine so significant for the location of factories?

In the early 1700s, British iron was of terrible quality. It was known as "pig iron"—it was brittle and snapped easily because it was full of impurities (carbon). If the Royal Navy wanted strong, flexible iron (known as "wrought iron") for its ship masts and cannons, it had to buy it from Sweden or Russia at a massive cost.

The man who solved this national crisis lived right here in Hampshire. Henry Cort was an ambitious navy agent who realized he could make a fortune if he could figure out how to mass-produce high-quality British iron. In 1775, he took over the **Funtley Ironworks** (just north of Fareham) because it was ideally located to supply the massive Royal Navy dockyards in nearby Portsmouth. At Funtley, Cort invented two revolutionary processes in 1783 and 1784: 1. **The Puddling Process:** He invented a new type of furnace where molten iron was stirred (or "puddled") with long iron rods to burn off carbon impurities. 2. **The Rolling Mill:** Instead of hammering the iron into shape by hand, Cort passed the red-hot iron through heavy, grooved rollers to squeeze out the last impurities and shape it into rails and bars.

KNOWLEDGE CHECK (Q5)

What was the primary purpose of the 'Puddling Process' invented by Cort?

- ☐ A. To hammer the iron into shape by hand.
- ☐ B. To stir molten iron and burn off carbon impurities.
- ☐ C. To make iron brittle and easy to snap.
- ☐ D. To import iron from Sweden more efficiently.

Q2. Why did Henry Cort decide to take over the Funtley Ironworks in 1775?

Q3. What two major inventions did Henry Cort patent at Funtley in the 1780s?

Q4. Study Source A. How does the physical design of the puddling furnace show that Cort was trying to solve the problem of 'pig iron' impurities?

Cort's inventions at Funtley changed the world. They allowed British iron production to multiply by 400% over the next twenty years, earning Cort the nickname "the father of the iron trade." The rails for the new train networks and the iron for the ships that built the British Empire were made using Cort's methods. Tragically, Cort did not die a wealthy hero. His business partner, Adam Jellicoe, had secretly stolen the money used to fund the Funtley Ironworks from the Royal Navy. When Jellicoe died, the government seized Cort's patents and ruined him. Cort died bankrupt and in poverty, but his local ingenuity powered a global empire.

Q6. Why did Cort die in poverty despite his world-changing inventions?

Q7. Why were Henry Cort's inventions so vital to the growth of the British Empire?

When assessing why the Funtley Ironworks was so successful in kickstarting the Industrial Revolution's iron industry, historians debate two main factors: **1. Geographical Luck (Location):** Funtley was situated just a few miles from the Portsmouth Royal Navy dockyards. In the 18th century, before the invention of trains, transporting heavy iron overland was incredibly expensive. Being located right next door to the largest industrial buyer of iron in the world (the Royal Navy) meant Cort had a massive, guaranteed market for his goods. Geography provided the economic opportunity. **2. Technological Genius (Innovation):** However, location alone wasn't enough. The Royal Navy desperately needed *wrought* iron (flexible and strong) for its ships and cannons, not the brittle British *pig iron* that snapped easily. Without Cort's brilliant inventions—the Puddling Furnace to burn off impurities and the Rolling Mill to speed up shaping—he would have had nothing useful to sell to the Navy. Innovation provided the solution to a national crisis. Historians must weigh these two factors to decide which was *more* important to his success.

Q8. Write a structured paragraph answering the following: "To what extent was the proximity to the Portsmouth dockyards the most important factor in the success of the Funtley Ironworks?"

Q9. Chronology Challenge: Place the following in the correct causal order and write one sentence explaining how each led to the next: Watt's Steam Engine, The Domestic System, Cort's Puddling Furnace.

L2: Was industrial work progress or punishment? (Factory conditions, child labor; Case Study: Funtley Clay Pits & Fareham Brickfields)

LEARNING OBJECTIVES

- [ ] Understand the core themes of Was industrial work progress or punishment? (Factory conditions, child labor; Case Study: Funtley Clay Pits & Fareham Brickfields).

Do Now Activity

1. Recall question from previous lesson.

Vocabulary Check

Write a historically accurate sentence connecting two terms from the glossary box below.

Child Labour | Brickfields

Your Sentence:

Source A: Fareham Brickfields

Source

Workers at a traditional brickfield in Fareham in the 19th century.

Placeholder narrative.

Q1. Placeholder task.

L3: Did industrialisation make British towns unlivable? (Urbanisation, Cholera, Public Health; Case Study: Fareham Red Bricks built Portsmouth & London)

LEARNING OBJECTIVES

- [ ] Understand the core themes of Did industrialisation make British towns unlivable? (Urbanisation, Cholera, Public Health; Case Study: Fareham Red Bricks built Portsmouth & London).

Do Now Activity

1. Recall question from previous lesson.

Vocabulary Check

Write a clear definition and a historically accurate sentence for the term: ***Urbanisation***

Source A: A Victorian Slum

Source

Overcrowded and dirty streets in a 19th-century British industrial town.

Placeholder narrative.

[[SRC_MARKER:L2_Task_0_0]] **Q1. Placeholder task.**

L4: How was the British Empire built and sustained? (Trade, Raj in India, Naval supremacy; Case Study: Portsmouth Dockyard & Royal Navy)

LEARNING OBJECTIVES

- [ ] Understand the core themes of How was the British Empire built and sustained? (Trade, Raj in India, Naval supremacy; Case Study: Portsmouth Dockyard & Royal Navy).

Do Now Activity

1. Recall question from previous lesson.

Vocabulary Check

Write a short paragraph using at least FOUR of the vocabulary words below correctly.

British Raj | Naval Supremacy

Source A: Portsmouth Dockyard

Source

HMS Victory and other Royal Navy ships at Portsmouth Dockyard, the heart of British naval power.

Placeholder narrative.

Q1. Placeholder task.

L5: How did ordinary people fight for a voice? (Peterloo, Chartism, Trade Unions; Case Study: Hampshire Swing Riots)

LEARNING OBJECTIVES

- [ ] Understand the core themes of How did ordinary people fight for a voice? (Peterloo, Chartism, Trade Unions; Case Study: Hampshire Swing Riots).

Do Now Activity

1. Recall question from previous lesson.

Vocabulary Check

Write a historically accurate sentence connecting two terms from the glossary box below.

Chartism | Trade Unions

Your Sentence:

Source A : The Swing Riots

Source

An illustration of agricultural workers protesting against the introduction of threshing machines during the Swing Riots.

Placeholder narrative.

Q1. Placeholder task.

L6: How did the road to democracy expand? (Reform Acts 1832–1884, Secret Ballot; Case Study: Hampshire MPs & Pocket Boroughs)

LEARNING OBJECTIVES

- [ ] Understand the core themes of How did the road to democracy expand? (Reform Acts 1832–1884, Secret Ballot; Case Study: Hampshire MPs & Pocket Boroughs).

Do Now Activity

1. Recall question from previous lesson.

Vocabulary Check

Write a clear definition and a historically accurate sentence for the term: ***Reform Acts***

Source A: A Pocket Borough

Source

A political cartoon mocking the corruption of un-reformed electoral constituencies.

Placeholder narrative.

Q1. Placeholder task.

L7: Who party truly benefited from 19th-century transformation? (Synthesis essay balancing economic growth vs. exploitation)

LEARNING OBJECTIVES

- [ ] Understand the core themes of Who party truly benefited from 19th-century transformation? (Synthesis essay balancing economic growth vs. exploitation).

Do Now Activity

1. Recall question from previous lesson.

Vocabulary Check

Write a short paragraph using at least FOUR of the vocabulary words below correctly.

Synthesis | Exploitation

Source A: The Two Nations

Source

An illustration showing the stark divide between the rich industrialists and the poor working classes in Victorian Britain.

Placeholder narrative.

[[SRC_MARKER:L6_Task_0_0]] **Q1. Placeholder task.**